

A Deep Dive into Scientific Writing with LaTeX

Physics and Mathematics Handbook

Author: Subhash

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Abstract

This handbook is designed as a practical LaTeX project that illustrates key concepts in algebra, calculus, physics, diagrams, tables, and references. It is structured to help learners build a five-page document with a title page, table of contents, equations, and illustrative diagrams.

A sample learning project for intermediate LaTeX users.

Contents

1 Algebra

1.1 Quadratic Formula

The quadratic formula solves $ax^2 + bx + c = 0$ with

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}. \quad (1)$$

For example, if $a = 1$, $b = 0$, and $c = -4$, then

$$x = \frac{0 \pm \sqrt{16}}{2} = \pm 2. \quad (2)$$

1.2 Binomial Theorem

The binomial theorem expands powers of sums:

$$(x + y)^n = \sum_{k=0}^n \binom{n}{k} x^{n-k} y^k. \quad (3)$$

For $n = 2$:

$$(x + y)^2 = x^2 + 2xy + y^2. \quad (4)$$

1.3 System of Equations

A linear system can be written as:

$$2x + 3y = 5, \quad (5)$$

$$4x - y = 11. \quad (6)$$

Solving by elimination gives:

$$2(2x + 3y) = 10, \quad (7)$$

$$4x - y = 11, \quad (8)$$

so $4x + 6y = 10$ and subtracting yields $7y = -1$, therefore

$$y = -\frac{1}{7}, \quad x = \frac{33}{7}. \quad (9)$$

2 Calculus

2.1 Derivatives

The derivative of a function measures its instantaneous rate of change. For example:

$$\frac{d}{dx} (x^3 - 5x^2 + 4x) = 3x^2 - 10x + 4. \quad (10)$$

2.2 Integrals

The definite integral computes signed area under a curve:

$$\int_0^2 (3x^2 - 10x + 4) dx = [x^3 - 5x^2 + 4x]_0^2 = -4. \quad (11)$$

2.3 Limits

A limit describes behavior as a variable approaches a value:

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1. \quad (12)$$

2.4 Taylor Series

The Taylor series for e^x around $x = 0$ is:

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots. \quad (13)$$

3 Physics

3.1 Newton's Laws

Newton's second law is expressed as

$$\mathbf{F} = m\mathbf{a}. \quad (14)$$

3.2 Kinetic Energy

The kinetic energy of a mass m moving at velocity v is

$$KE = \frac{1}{2}mv^2. \quad (15)$$

For $m = 4 \text{ kg}$ and $v = 6 \text{ m/s}$:

$$KE = \frac{1}{2}(4)(6)^2 = 72 \text{ J}. \quad (16)$$

3.3 Momentum

Linear momentum is defined by

$$\mathbf{p} = m\mathbf{v}. \quad (17)$$

3.4 Ohm's Law

Ohm's law relates voltage, current, and resistance:

$$V = IR. \quad (18)$$

If $V_s = 12.0 \text{ V}$ and $R = 6.0 \Omega$, then

$$I = \frac{V_s}{R} = 2.00 \text{ A}. \quad (19)$$

3.5 Wave Equation

The one-dimensional wave equation is

$$\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}. \quad (20)$$

Table 1: Common Physics Quantities

Quantity	Symbol	Typical Unit
Mass	m	kg
Velocity	v	m/s
Force	F	N
Energy	E	J
Resistance	R	Ω

4 Diagrams and Visualization

4.1 Coordinate System and Parabola

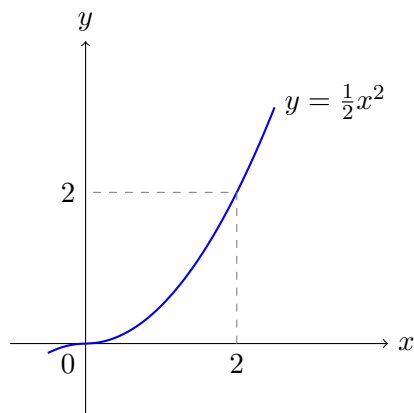


Figure 1: Graph of a simple parabola in a coordinate system.

5 Conclusion

This handbook demonstrates how LaTeX can produce structured academic content with equations, tables, graphics, and references. Progressing through these examples gives a strong foundation for creating longer technical documents.

References

References

- [1] Leslie Lamport, *LaTeX: A Document Preparation System*, 2nd edition, Addison-Wesley, 1994.
- [2] Donald E. Knuth, *The TeXbook*, Addison-Wesley, 1984.